



Changes in aggregation properties and the metabolite production of probiotics following treatment with polysaccharides derived from the edible mushroom *Cordyceps militaris*

Ji Young Kang ^a , Seonghun Kim ^{a b}, Jung-Mi Kim ^c

[Show more](#)

Outline | Share Cite

<https://doi.org/10.1016/j.lwt.2024.116845>

[Get rights and content](#)

Under a Creative Commons [license](#)

[Open access](#)

Highlights

- LCMP is a polysaccharide composed of mannose and glucose in a 3.5:1 ratio.
- Isolated LAB strains exhibited auto-aggregation in low-glucose medium with LCMP.
- The LAB strains inhibited the growth of pathogenic bacteria.
- LCMP stimulates the LAB to produce more lactic acid and other organic acids.
- The LAB grown with LCMP can effectively inhibit the proliferation of HT-29 cells.

Abstract

Low-molecular-weight exopolysaccharides of *Cordyceps militaris* (LCMP), obtained by enzymatically digesting the exopolysaccharide produced by *C. militaris*, were purified and found to be comprised of 77.9% mannose and 22.1% glucose. Additionally, lactic acid bacteria (LAB) strains, including *Enterococcus faecalis* strain L4, *Pediococcus acidilactici* strain L25, and *Companilactobacillus zhachilii* strain L113, were isolated and their growth phenotypes and metabolites appeared in the presence or absence of purified LCMP were determined. In medium containing purified LCMP with low glucose, strains L4, L25, and L113 all exhibited auto-aggregation; however, co-aggregation occurred only with the strain L113. Moreover, culture media of strains L4, L25, and L113 in either the presence or the absence of LCMP could all inhibit the growth of pathogenic bacteria. Culture media of strains L4, L25, and L113 with purified LCMP contained higher concentrations of lactic acid and other organic acids than in media cultured without purified LCMP. Additionally, the proliferation of HT-29 cells was more effectively inhibited by culture media of strains L4, L25, and L113 with purified LCMP. Together, these results indicated that LCMP may provide positive effects for the host by promoting probiotics to inhibit the growth of pathogens and increase the production of metabolites related to anticancer activity.



Keywords

Exopolysaccharides (EPS); Lactic acid bacteria (LAB); Metabolites; Aggregation; Anticancer activity

1. Introduction

Cordyceps militaris is an edible mushroom used as traditional medicine and a health food in Asian countries. Over the past decade, research has shown that polysaccharides in this mushroom have multiple biological activities, including antioxidant and anticancer activities (Chen et al., 2021; Mwangi, Macharia, Wagara, & Bence, 2022; Rathore, Prasad, & Sharma, 2017). In addition, the polysaccharides have been shown to affect the intestinal microbial community, thereby indirectly modulating the intestinal health of the host (Chen et al., 2021; Mao et al., 2020). When eaten as a food or supplement, the mushroom and its extracted polysaccharides can directly affect the host's intestinal microbial community or intestinal cells and/or be digested or modified by intestinal microbial enzymes in the host intestine (Maske et al., 2021).

Pullulanase is an enzyme produced by intestinal bacteria; it is an α -amylase that belongs to glycoside hydrolase families 13 and 57 (Nisha & Satyanarayana, 2013). Pullulanase hydrolyzes α -1,6-linked branches in polysaccharides, including glycogen, amylopectin, and pullulan. Pullulanase has

been identified in a broad range of bacteria, including *Streptococci*, *Bacillus*, and *Lactobacillus* (Hytönen, Haataja, & Finne, 2003; Møller et al., 2017; Rahmatabadi et al., 2017). A previous study suggested that pullulanases from gut bacteria are important to other bacteria for the utilization of branched malto-oligosaccharides in the gastrointestinal tract (GIT) (Møller et al., 2017).

Lactic acid bacteria (LAB) are Gram-positive anaerobic or facultative aerobic cocci or rods that colonize the intestine and include *Lactobacillus*, *Enterococcus*, *Bifidobacterium*, and *Streptococcus* species. LAB produce lactic acid as one of the main fermentation products of carbohydrate metabolism (Françoise, 2010; Liu, Pang, Zhang, & Cai, 2014) and are among the strains most commonly used as potential probiotics because they benefit the host by maintaining colonic health and preventing colorectal cancer and other gut-related diseases (Kumar & Sharma, 2022; Masood, Qadir, Shirazi, & Khan, 2011; Uccello et al., 2012). However, some strains of *Enterococcus* and *Streptococcus* genera are also known to be opportunistic pathogens that can occasionally cause disease (Ramos, Silva, Dapkevicius, Igrejas, & Poeta, 2020; Stewart, 2022). The normal intestinal microbiota essentially starves invading pathogens by consuming the limited resources commonly required for the growth of competing pathogens. Commensal bacteria can also produce specific metabolites or active substances that affect the virulence of pathogens and inhibit their growth (Debnath, Kumar, Kumar, Mehta, & Yadav, 2021; Kamada, Chen, Inohara, & Núñez, 2013). In this way, the intestinal microbiota maintains a delicate balance. Disruption of the balance triggers dysbiosis, pathogen overgrowth, and eventual pathologic immune responses and disease (Chidambaram et al., 2022; Chu et al., 2023; Kamada et al., 2013).

Postbiotics are the active substances produced by probiotics and include soluble factors and the repertoire of metabolites produced by the fermentation of dietary components (Johannsen & Prescott, 2009; Peluzio, Martinez, & Milagro, 2021). Thus, the production and composition of these postbiotics in the gut are directly impacted by host dietary intake and the gut microbiota. In addition, because different bacterial species have distinct fermentation pathways, the diversity and quantity of fiber in a diet induce the production of differing amounts and kinds of metabolites by these bacteria, such as short-chain fatty acids (SCFAs) (Cronin, Joyce, O'Toole, & O'Connor, 2021; Gold & Zhu, 2022). Moreover, metabolite production by bacteria is strain specific, thus the profiles of individual probiotic species and strains should be considered when developing prebiotics for host health (Chung et al., 2016).

In a previous study, we prepared exopolysaccharides from the edible mushroom *C. militaris* (CMP) and obtained low-molecular-weight exopolysaccharides of *C. militaris* (LCMP) by pullulanase digestion. We demonstrated that crude LCMP had antioxidant activity and were excellent prebiotics that decreased Firmicutes/Bacteroidetes ratios and increased beneficial metabolites, such as lactic acid, acetic acid, and phenolic acid, in the gut microbiota consortium (Kang, Lee, Kim, Choi, & Kim, 2022). In the present study, we aimed to investigate how the LCMP affects probiotic bacteria within the consortium. We first purified crude LCMP by ion-exchange chromatography and size-exclusion chromatography. To understand the interactions of purified LCMP and probiotic strains, LAB strains

isolated from swine gut microflora were cultured in the presence or absence of purified LCMP and then their growth phenotypes and antimicrobial activities were determined. In addition, metabolites in the media of LAB strains cultured in the presence of purified LCMP were analyzed, and their anti-proliferation activity against human colon cancer cells was evaluated.

2. Materials and methods

2.1. Preparation and purification of prebiotic LCMP

To obtain LCMP, the exopolysaccharide of *C. militaris* (CMP) was digested with pullulanase (Sigma, Burlington, MA) and autoclaved to eliminate the enzyme activity as previously described ([Kang et al., 2022](#)). The crude LCMP was purified by a two-step process. First, crude LCMP was applied to a column with DEAE-52 resin (Biophoretics, Sparks, NV) and eluted with 0.2–1.0 M NaCl (0.2 M intervals) in pH 6.0 phosphate buffer at a flow rate of 0.5 ml/min. The major fractions were then applied to a column with Sephacryl S-500 resin (Cytiva, Marlborough, MA) and eluted with water at a flow rate of 0.5 ml/min. The total carbohydrate content of the fractions was measured according to the phenol-sulfuric acid method ([Masuko et al., 2005](#)) with glucose as the standard. To compare polysaccharide sizes, blue dextran (average MW 2,000,000 Da, Sigma, Burlington, MA) and dextran sodium sulfate (MW 6500–10,000 Da, Sigma, Burlington, MA) were also used.

2.2. Monosaccharide composition of polysaccharides

The monosaccharide composition analysis was performed using high-performance liquid chromatography (HPLC, Agilent 1260) with an Aminex HPX-87P column (BioRad, Hercules, CA) and 0.005 N H₂SO₄ as the mobile phase. To hydrolyze the polysaccharides, trifluoroacetic acid (TFA, Sigma, Burlington, MA) was added to the purified fraction for 5 h at 100 °C until reaching the final concentration of 2 M. Next, TFA was removed by rotary evaporation and the dried sample was dissolved in ultrapure water. The aqueous sample was filtered using a 0.2 µm syringe filter before HPLC analysis. Glucose and mannose were used as the reference controls and were subjected to the analysis.

2.3. Bacterial strain and media

LAB isolates were obtained from pre-cultured swine gut microbiota samples grown in MRS medium (Difco, Franklin Lakes, NJ) containing LCMP under anaerobic conditions at 37 °C for 5 d. To identify the strains, their 16S rRNA gene sequences were amplified from purified genomic DNA using universal primers (27F and 1492R). The pairwise gene similarity of the 16S rRNA sequences was determined using the EzBioCloud database ([Yoon et al., 2017](#)) and the BLAST program. A phylogenetic tree was then constructed using methods described in a previous study ([Kang, Song, & Kim, 2023](#)). Pathogenic bacterial strains Staphylococcus aureus subsp. *aureus* (KCTC 2881), Bacillus cereus (KCTC 1014), Salmonella enterica subsp. *enterica* (KCTC 2421), and Escherichia coli (KCTC

2441^T) were obtained from the Korean Collection for Type Culture (KCTC, Korea). *S. enterica* was grown on nutrient agar (NA, Difco, Franklin Lakes, NJ) containing 0.5% NaCl, and the other pathogenic bacteria were grown on NA at 37 °C for 1 d.

2.4. Acid and bile resistance of LAB strains

Acid and bile resistance of LAB strains were determined by the viable plate count method. Exponential phase cells were collected by centrifugation and adjusted to an OD₆₀₀ of 0.5 with 10 mM sterile phosphate buffer pH 7.0. Then, 500 µL of the bacterial suspensions were added to 5 mL of MRS medium pH 2.0 or 3.0 for the acid resistance test. To test bile resistance, suspensions were added to 5 mL of MRS medium containing 0.1% or 0.3% bile salt (Sigma, Burlington, MA). These cultures were incubated at 37 °C for 3 h, diluted, and spread onto MRS agar. After 2 d, colonies were counted and survival was calculated using the formula: $N_{3h}/N_{0h} \times 100$ in which N_{3h} and N_{0h} represent the number of colonies after 3 h of incubation and before adding acidic and bile salt media, respectively.

2.5. Antimicrobial activity of LAB strains

The antibacterial activities of the LAB strains were determined by evaluating their effects on the growth of pathogenic bacteria. LAB strains were cultured in MRS medium supplemented with or without a final concentration of 1% purified LCMP at 37 °C for 48 h under anaerobic conditions, and then centrifuged at 8000 rpm for 10 min. The supernatants were collected and filtered through 0.22 µm filters. The pathogenic strains were spread onto NA or NA containing 0.5% NaCl plates, 100 µL of filtered supernatants were spotted onto the plates ($\varnothing = 5$ mm), and the plates were incubated at 37 °C for 24 h. Diameters of the zones of inhibition were measured in millimeter (mm).

2.6. Auto- and co-aggregation assays

To test auto-aggregation, LAB strains were grown in 2 mL of the indicated medium at 37 °C for 24 h under anaerobic conditions. Cells were harvested by centrifugation at 5000 g for 20 min at 4 °C, washed twice with phosphate-buffered saline (PBS, pH 7.0), and resuspended in the same buffer. Suspensions were vortexed, incubated at 37 °C for 2 h, and then absorbances at 600 nm were measured. The percent aggregation was calculated using the formula: $1 - (A_{2h}/A_0) \times 100$ in which A_0 and A_{2h} represent absorbances at the initial time and 2 h, respectively.

To test co-aggregation of the LAB strains with the pathogenic bacteria strains, bacterial suspensions were prepared as described above. Then, equal volumes (500 µL) of the various LAB strain suspensions and the various pathogenic strain suspensions were mixed 1:1, incubated at 37 °C for 2 h without agitation, and absorbances at 600 nm were measured. The percent co-aggregation was calculated using the formula: $(A_{path} + A_{LAB})/2 - (A_{mix})/(A_{path} + A_{LAB})/2 \times 100$ in which A_{path} and A_{LAB} represent the absorbance values of the control tubes containing only the pathogen and only the LAB strain, respectively, and A_{mix} represents the absorbance value of the mixed suspension after 2 h of

incubation.

2.7. Metabolite extraction and analysis

To extract secreted metabolites, LAB strains were grown in MRS medium supplemented with or without a final concentration of 1% purified LCMP at 37 °C for 72 h under anaerobic conditions. The cultures were centrifuged at 8000 rpm for 20 min to obtain the cell-free supernatant. The supernatant was transferred to a new tube and acidified to pH2 by the addition of HCl. This acidified supernatant (1 ml) was mixed with an equal volume of diethyl ether, the samples were centrifuged, the organic solvent was transferred to a new tube, and residual water in the organic solvent was removed by supplementation with Na₂SO₄. The organic solvent containing metabolites (200 µl) was transferred into gas chromatography (GC) vials and 10 µl of N,O-Bis(trimethylsilyl)trifluoroacetamide (Sigma, Burlington, MA) was added for derivatization at 25 °C for 18 h. Metabolite analysis was performed by GC-mass spectrometry (MS) using an Agilent 7000 GC-MS machine equipped with a VF-5ms column (30 m × 0.25 mm diameter × 0.25 mm film thickness; Agilent, Santa Clara, CA) as described previously ([Kang et al., 2022](#)).

2.8. Human colon cancer cell culture and proliferation assay

The postbiotic anti-proliferation effects of LAB strain culture media were investigated using human colon adenocarcinoma cell line HT-29. HT-29 cells were cultured as monolayers in 9 cm culture dishes in Dulbecco's modified Eagle's medium (DMEM, Gibco, Waltham, MA) supplemented with 10% fetal bovine serum (Invitrogen, Waltham, MA), 1% L-glutamine (v/v) and 1% penicillin/streptomycin and maintained at 37 °C in a humidified atmosphere of 5% CO₂. HT-29 cells were sub-cultured following enzymatic digestion with 1% trypsin. To measure HT-29 cell proliferation, 1 × 10⁴ cells in 200 µL of DMEM were seeded into each well of a 96-well microtiter plate and incubated for 24 h. After washing with PBS twice, HT-29 cells were treated with 5% (v/v) and 15% (v/v) LAB strain culture media for 1–3 d at 37 °C. The LAB strain culture media were obtained by passing it through a 0.2-µm syringe filter after cultivation with or without purified LCMP (final concentration, 1%). As a negative control, the cells were treated with MRS medium at the same concentrations. Then, HT-29 cells were washed with PBS, incubated for 3 h with 20 µL of MTS (CellTiter 96 AQueous One solution, Promega, Madison, WI), and absorbances at 490 nm were measured using a microplate reader (BioTek Inc., Winooski, VT).

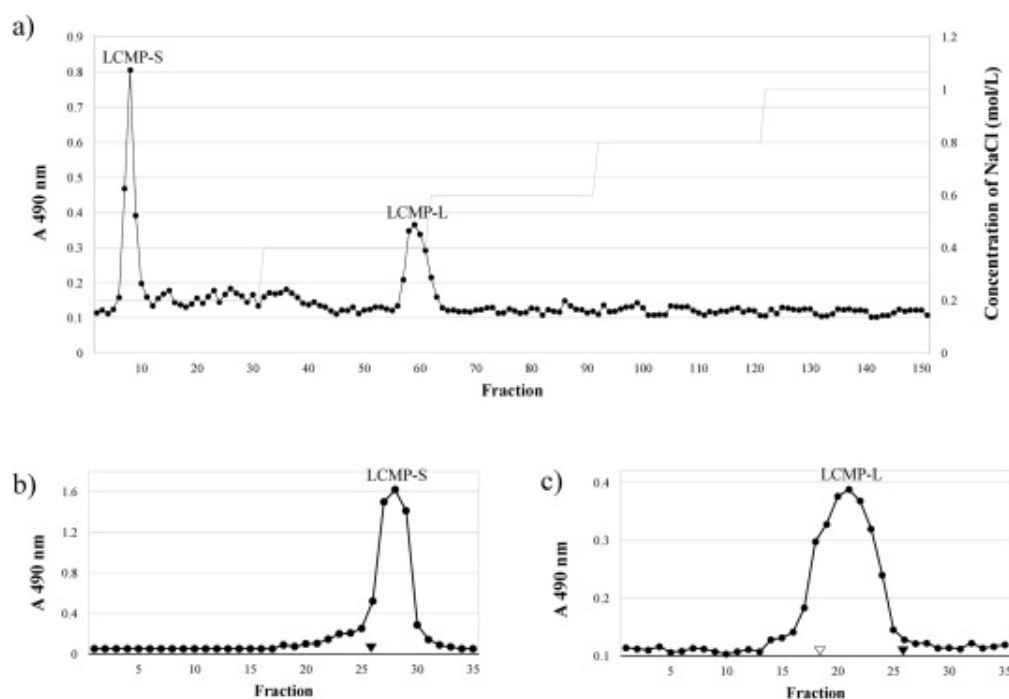
2.9. Statistics

Each experiment was independently replicated three times. To statistically evaluate differences between groups, two-way paired Student *t*-tests were performed using the software Microsoft Excel (Microsoft, Redmond, WA); *p* < 0.05 was considered statistically significantly different.

3. Results and discussion

3.1. Purification and monosaccharide composition of LCMP

In a previous study, exopolysaccharides prepared from the edible mushroom *C. militaris* were digested enzymatically to produce LCMP, which was shown to have prebiotic and antioxidant activity. In this study, crude LCMP was purified with a DEAE-52 column to obtain two major polysaccharide fractions (Fig. 1a), which were further purified with a Sephacryl S-500 column. A single peak was observed in each fraction, and the fractions were designated LCMP-S and LCMP-L (Fig. 1b and c). The molecular weight of LCMP-S was estimated to be < 6500 Da indicating a monosaccharide or disaccharide branch chain from the backbone of CMP. Thin-layer chromatography analysis also suggested that LCMP-S is a monosaccharide or a disaccharide (Fig. S1). The molecular weight of LCMP-L was slightly smaller than blue dextran, which has a molecular weight of 2,000,000 Da (Fig. 1c). The viscosities of LCMP-L in fractions 18–23 were higher than in the other fractions, which is characteristic of a linear polysaccharide (Guo, Hu, Wang, & Ai, 2017). A previous study reported crude LCMP to be 1.6×10^3 kDa and more gelatinous due to its linear form resulting from reductions in branched links compared with CMP (Kang et al., 2022). Purified LCMP-L and crude LCMP had identical characteristics and consisted of only one polysaccharide, therefore, LCMP-L was named ‘purified LCMP’.



[Download: Download high-res image \(323KB\)](#)

[Download: Download full-size image](#)

Fig. 1. FPLC elution curves of LCMP on DEAE-52 and Sephacryl S-500 HR columns. a) Elution profile of crude LCMP on the DEAE-52 column. b) Elution profile of LCMP-S on the Sephacryl S-500 HR column. c) Elution profile of LCMP-L on the Sephacryl S-500 HR column. The white triangle indicates blue dextran (2,000,000 Da) and the black triangle indicates dextran sulfate (6500–10,000 Da).

To investigate differences between CMP and purified LCMP, their monosaccharide compositions were measured by HPLC after acid hydrolysis. In the previous study, pullulanase-treated CMP was separated into soluble and insoluble fractions, and the soluble fraction was designated as crude LCMP. CMP has been reported to consist 1:1.3 ratio of mannose to glucose. This result was similar to the values of 45.6% mannose and 54.5% glucose measured in this study. Interestingly, purified LCMP was comprised of 77.9% mannose and 22.1% glucose, which is approximately 3.5 times as much mannose than glucose ([Table 1](#)). Monosaccharide composition analysis of the insoluble fraction and LCMP-S showed that they consisted of glucose only ([Table 1](#) and [Figs. S1d and e](#)). Mannose-containing polysaccharides from several mushroom species have shown antioxidant, cytokine-stimulating, and anticancer effects ([Chen, Hsu, Lin, Lai, & Wu, 2006](#); [Kim et al., 2006](#); [Wang, Wang, Xu, & Ding, 2017](#); [Xiang, Xu, & Li, 2012](#)). Therefore, the high proportion of mannose in purified LCMP may contribute to its bioactive properties.

Table 1. Monosaccharide composition of polysaccharides derived from *C. militaris*. Results are presented as the average ± standard deviation of three experiments.

Polysaccharide				Molar Ratio (%)	
				Mannose	Glucose
CMP				45.6 ± 0.85	54.4 ± 0.85
CMP treated with pullulanse	Insoluble part			–	100
	Soluble part (crude LCMP)		LCMP-L	77.9 ± 0.65	22.1 ± 0.65
			LCMP-S	–	100

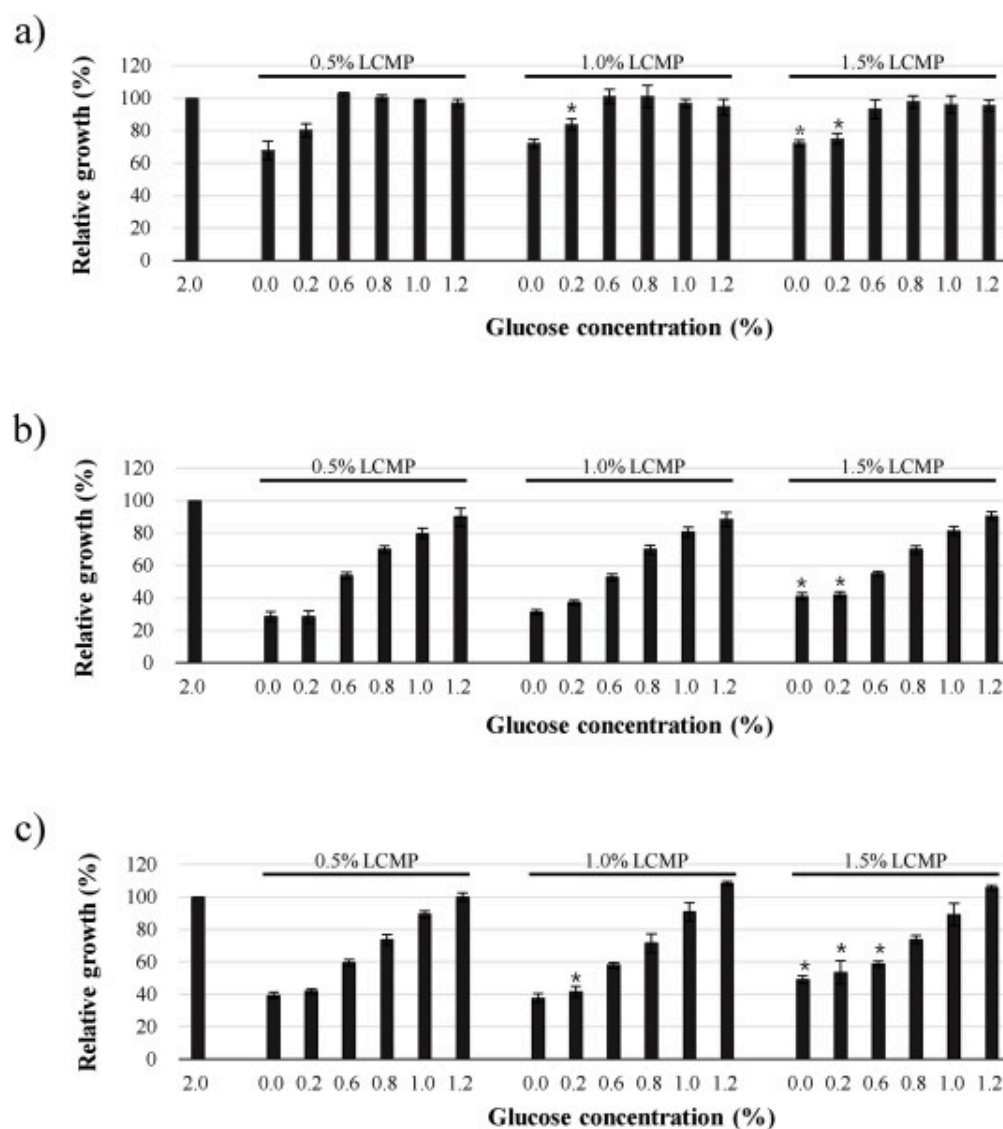
CMP, exopolysaccharides produced by *C. militaris* in submerged culture.

3.2. Screening and isolation of LAB strains

Because some characteristics of bacteria, such as mechanisms and triggers for the aggregation phenomenon or the production of metabolites, are species and/or strain specific ([Chung et al., 2016](#); [Isenring, Geirnaert, Lacroix, & Stevens, 2021](#)), we attempted to screen LAB strains that interacted with LCMP from gut microbiota samples containing complicated and diverse microorganisms. Gut microbiota samples were pre-cultured in MRS medium supplemented with purified LCMP instead of glucose as a carbon source for 3 d, and the cultures were then spread onto MRS agar plates containing purified LCMP and incubated at 37 °C under anaerobic conditions. Several colonies that presented after 5 d were purified by single colony isolation and identified by 16S rRNA gene sequencing. Among the identified colonies, three strains, L4, L25, and L113, were selected as representatives of their taxonomic groups and their growth characteristics. However, most of the other isolates showed no significant differences in the characteristics described below following

treatment with purified LCMP. A comparative analysis of the 16S *rRNA* genes showed that sequences of the strain L4, L25, and L113 were most similar to those of *Enterococcus faecalis* ATCC 19433^T, *Pediococcus acidilactici* DSM 20284^T, and *Companilactobacillus zhachilii* KCTC 21106^T, with sequence similarities of 99.9%, 99.9%, and 99.5%, respectively. The gene sequences were deposited in the *National Center for Biotechnology Information* (NCBI) GenBank database and are available through accession numbers PP843233, PP843234, and PP843235. The result of the *phylogenetic analysis* based on the 16S *rRNA* gene sequences are presented in Fig. S2.

To investigate the effects of purified LCMP on the growth of the isolated LAB strains, the strains were grown in MRS medium supplemented with purified LCMP instead of glucose as the carbon source. In media containing 0.5–1.5% purified LCMP as the sole carbon source, the LAB strains did not grow well. However, when the media containing purified LCMP was supplemented with 0.6, 0.8, 1.0, and 1.2% glucose (final concentrations), the LAB strains grew well (Fig. 2). This result indicates that purified LCMP is not a preferred carbon source for LAB strain growth. Therefore, in the further experiments to compare LAB strain characteristics during growth with purified LCMP, the strains were grown in 1% glucose and 1% purified LCMP unless otherwise indicated.



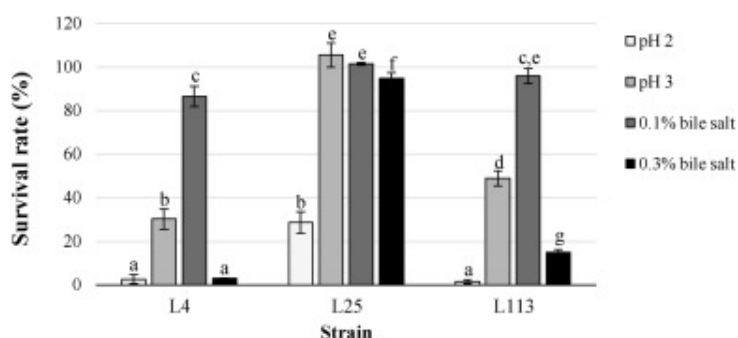
[Download: Download high-res image \(559KB\)](#)

[Download: Download full-size image](#)

Fig. 2. Relative growth of *E. faecalis* strain L4 (a), *P. acidilactici* strain L25 (b), and *C. zhachilii* strain L113 (c) in media containing various concentrations of purified LCMP and glucose. Growth was measured by determining the optical density at 600 nm (OD_{600}) after 3 d. The first bar in each graph represents general MRS medium (containing 2% glucose) without LCMP. Results are presented as averages ($n = 3$) with standard deviations. Asterisks indicate where the auto-aggregation phenotype is observed.

3.3. Acid and bile salt tolerance of LAB strains

To survive and produce postbiotics in the gastrointestinal system of the host, LAB strains must be acid and bile salt tolerant. To evaluate acid resistance of the isolated LAB strains, pre-cultured cells were incubated in MRS media at pH 2 and 3 for 3 h. *P. acidilactici* strain L25 showed superior survivability at pH 3 ($105.5 \pm 5.6\%$), whereas the survivabilities of *E. faecalis* strain L4 and *C. zhachilii* strain L113 were lower at $30.3 \pm 4.5\%$ and $48.9 \pm 3.4\%$, respectively. The survivabilities of all the strains were dramatically lower at pH 2 than at pH 3 (Fig. 3). To evaluate bile salt resistance, LAB strains were incubated in MRS medium with 0.1 and 0.3% (w/v) bile salt. All strains survived in MRS medium with 0.1% bile salt; the survival percentages of *E. faecalis* strain L4, *P. acidilactici* strain L25, and *C. zhachilii* strain L113 were $86.5 \pm 4.7\%$, $101.5 \pm 0.7\%$, and $96.0 \pm 3.4\%$, respectively. In MRS medium with 0.3% bile salt, *P. acidilactici* strain L25 had a high survival percentage of $94.8 \pm 2.7\%$, whereas the survival percentages of *E. faecalis* strain L4 and *C. zhachilii* strain L113 were low at $3.0 \pm 0.3\%$ and $15.1 \pm 1.3\%$, respectively (Fig. 3). Although their survivability in media at pH 2 and 0.3% bile salt was relatively low, because the LAB strains were isolated from gut microbial samples and because they were resistant to a low pH of 3, they may function in the intestine as probiotics. Therefore, these LAB strains were used as probiotics to investigate their interaction with purified LCMP as the prebiotic in this study.



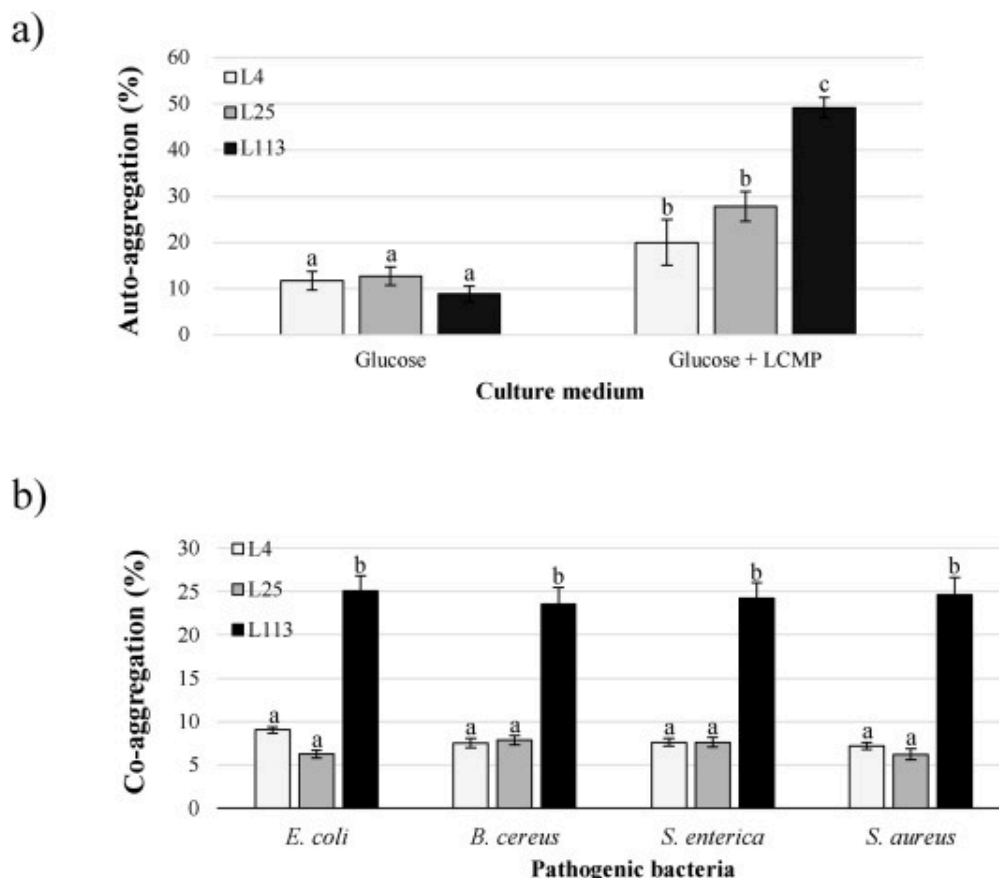
[Download: Download high-res image \(98KB\)](#)

[Download: Download full-size image](#)

Fig. 3. Comparative analysis of the survival of LAB strains under acidic and bile salt conditions. Results are presented as averages ($n = 3$). Bars with the same letter did not differ significantly ($p < 0.05$).

3.4. Auto- and co-aggregation of LAB strains

The auto-aggregation phenomenon of bacteria could facilitate cell-cell interaction and genetic exchange (Busch & Stumm, 1968; Choi & Woo, 2015; Trunk, Khalil, & Leo, 2018). In addition, the aggregation ability of probiotics has been correlated with adherence, colonization, and co-aggregation with pathogenic bacteria in the gut (Isenring et al., 2021; Nwoko & Okeke, 2021; Zawistowska-Rojek, Kośmider, Stępień, & Tyski, 2022). To investigate the auto-aggregation levels of the LAB strains, strains were cultured in media containing various concentrations of glucose and purified LCMP as the carbon sources. The auto-aggregation phenotype was observed in LAB strains grown in MRS medium containing purified LCMP instead of glucose but not in MRS medium without supplementation (Fig. S3). The auto-aggregation rate was highest in *C. zhachilii* strain L113 (49.2%), followed by *P. acidilactici* strain L25 (27.8%) and *E. faecalis* strain L4 (20.0%) (Fig. 4a). The auto-aggregation phenotype was observed in medium with a high concentration of purified LCMP and a low concentration of glucose (Fig. 2, asterisks) but not in medium with a low concentration of purified LCMP (<1.0%) or a high concentration of glucose (>0.6%).



[Download: Download high-res image \(322KB\)](#)

[Download: Download full-size image](#)

Fig. 4. Aggregation of LAB strains cultured in media with or without purified LCMP. Auto-aggregation (a) and co-aggregation (b) of LAB strains were measured after 2 h of incubation in PBS. Results are presented as averages (n = 3). Bars with the same letter did not differ significantly

($p < 0.05$).

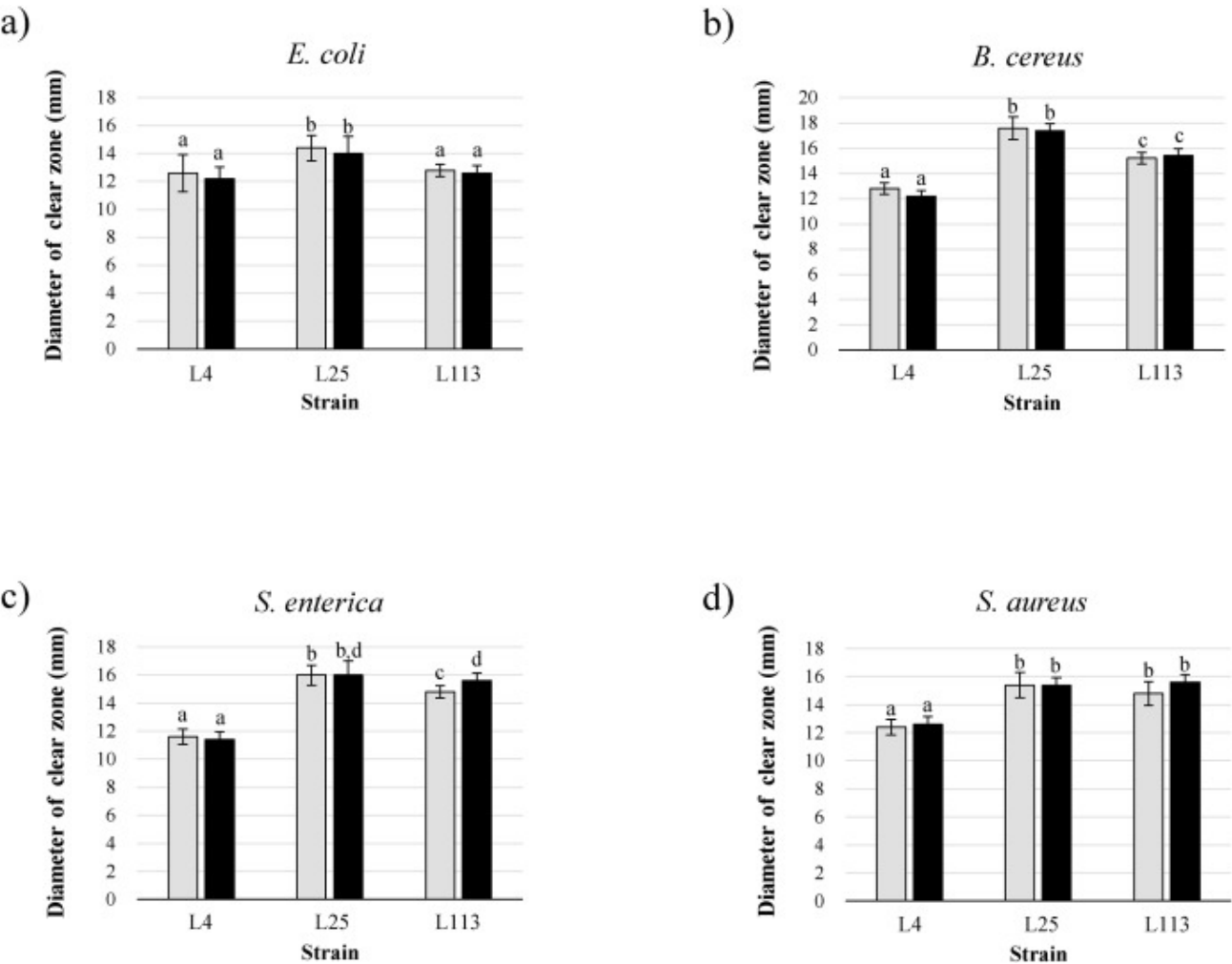
C. zhachilii strain L113, which displayed the most auto-aggregation, also co-aggregated with all of the tested pathogenic bacteria (*S. aureus* subsp. *aureus*, *B. cereus*, *S. enterica* subsp. *enterica*, and *E. coli*). However, the other LAB strains did not co-aggregate significantly with pathogenic bacteria (Fig. 4b). Therefore, *C. zhachilii* strain L113 may disturb adhesion and/or growth of pathogenic bacteria by co-aggregation in the presence of purified LCMP in the gut.

A peculiarity of co-aggregation between cells of different microbial species is known to be related to the ability of probiotic bacteria to attach to pathogenic bacteria and thereby inhibit adhesion of the pathogens to mucosal cells in the gastrointestinal tract (Hojjati, Behabehani, & Falah, 2020; Juntarachot et al., 2023; Zawistowska-Rojek et al., 2022). In *Lactobacillus*, the adhesion property has been reported to be associated with the presence of surface proteins, including sortase-dependent protein, mannose-binding protein, mucus-binding protein, and collagen-binding protein (Gross et al., 2008; Malik et al., 2013; Singh, Kumar, Mohanty, Grover, & Kaushik, 2018; Yadav et al., 2013). Besides, starvation of *L. plantarum* strains isolated from the gut enhanced auto-aggregation, but re-cultivation under standard conditions inhibited aggregation. However, wild-type and non-aggregating *L. plantarum* strains did not display the auto-aggregation phenotype under starvation conditions indicating that auto-aggregation is strain specific characteristics (Isenring et al., 2021). These results imply that the abundant mannose present in purified LCMP may induce the increased expression of cell surface proteins, such as the mannose-binding protein in *C. zhachilii* strain L113, isolated from gut microbiota samples in this study, thereby increasing auto- and co-aggregation. Consequently, aggregation induced by the LCMP can help strains adhere to epithelial cells, which would increase their survival. Furthermore, the surviving strains may produce beneficial substances that can aid in maintaining intestinal balance.

3.5. Antimicrobial activity of LAB strains

To evaluate the antibacterial activities of the isolated LAB strains, the diameters of the zones of inhibition against intestinal pathogenic bacteria (*S. aureus* subsp. *aureus*, *B. cereus*, *S. enterica* subsp. *enterica*, and *E. coli*) were measured. *P. acidilactici* strain L25 displayed antimicrobial activity against all the tested pathogenic bacteria with the largest zones of inhibition ranging from 14.4 to 17.6 mm (gray bars) or 14.0–17.4 mm (black bars) in the absence or presence, respectively, of purified LCMP in the growth medium. *E. faecalis* strain L4 and *C. zhachilii* strain L113 also displayed antimicrobial activity against all of the tested pathogenic bacteria, with zones of inhibition ranging respectively from 11.6 to 12.8 mm or 11.4–12.6 mm and 12.8–15.2 mm or 12.6–15.6 mm in the absence (gray bars) or presence (black bars) of purified LCMP in the growth medium (Fig. 5). These results show that the antibacterial activities of the LAB strains were not affected by the presence of purified LCMP in the media (Fig. 5). Rather, the degree of antimicrobial activity seemed to depend on the degree of the cell growth of each strain because there was no antimicrobial activity observed in cultures containing 0.2% glucose or only purified LCMP without glucose (data not shown), which

showed relatively low growth rates. In addition, the pH of the culture media after cultivation with purified LCMP was lower than the pH of the culture media after cultivation without purified LCMP (Table 2). Although the pH values of these culture media differed, their antibacterial activities were almost the same for each pathogen (see Fig. 5, gray and black bars). These results indicate that antibacterial activity was not affected by the presence of purified LCMP or the low pH value in the medium and may be due to the presence of antibacterial metabolites, such as bacteriocins, produced by the strains.



[Download: Download high-res image \(387KB\)](#)
[Download: Download full-size image](#)

Fig. 5. Antimicrobial activities of LAB strains were measured as diameters of the zones of inhibition. LAB strains were cultured in the absence (gray bars) or presence (black bars) of purified LCMP, and the cell-free cultured media were assessed for antimicrobial activity. Results are presented as averages (n = 3). Bars with the same letter did not differ significantly (p < 0.05).

Table 2. pH values and relative abundances of metabolites in LAB strain culture media. ‘+’ represent relative abundances and ‘-’ indicates not detected from three experiments.

Strains	Carbon	pH	Retention time of metabolites (min)
---------	--------	----	-------------------------------------

source ^a				5.5	7.5	9.7	10.3	15.7	18.0	18.9
				Lactic acid (+TMS)	Lactic acid (+2TMS)	2-Hydroxy isocaproic acid	Benzoic acid	3-Phenyl lactic acid	Palmitic acid	Stearic acid
L4	Glucose	4.87 ± 0.23	+		–	–	–	–	–	–
	+ LCMP	4.23 ± 0.21	++		+++	+	–	–	++	+
L25	Glucose	4.68 ± 0.29	+		++	–	–	–	–	–
	+ LCMP	4.17 ± 0.23	+		+++	–	++	++	+	–
L113	Glucose	4.59 ± 0.15	++		–	–	–	–	–	–
	+ LCMP	4.32 ± 0.18	+++		++	–	+	–	–	–

TMS, trimethylsilyl derivatives.

a
Cultured media supplemented with only glucose as the carbon source (Glucose) or glucose with purified LCMP (+LCMP).

Bacteriocins produced by LAB are ribosomal-synthesized antimicrobial peptides and have been approved as food additives for natural antimicrobials ([Sugrue, Ross, & Hill, 2024](#)). In recent years, bacteriocins have been demonstrated to have anticancer effects on cancer cells ([Goh et al., 2022](#)). The properties of bacteriocins, such as pediocin from Pediococcus spp. and enterocin from Enterococcus spp., have been well characterized ([Kasimin et al., 2022](#); [Papagianni & Anastasiadou, 2009](#)). In a previous study, bacteriocin produced by *P. acidilactici* strain ST3522BG isolated from silage samples showed the antimicrobial activities against Listeria monocytogenes, *L. innocua* and Enterococcus strains previously characterized as vancomycin-resistant ([Fugaban et al., 2022](#)). A cell-free supernatant produced by *P. acidilactici* strain LWX 401 isolated from pickles was recently reported to have antibacterial activity against foodborne pathogens such as *E. coli* O157:H7, *S. aureus*, *S. agalactiae*, *S. choleraesuis*, *S. para-typhi* type B, and *S. enteritidis*. Furthermore, metabolomics analysis revealed that the supernatant contained metabolites of mostly lipids (25.3%) and organic acids (23.3%). The authors proposed that the lipids and lipid-like molecules of the metabolites produced by *P. acidilactici* strain LWX 401 may exert the antibacterial activities at least partly by interacting with phospholipids within the pathogenic bacterial membrane ([Liu et al., 2024](#)).

Although *E. faecalis* is considered to be an opportunistic pathogen, bacteriocin or bacteriocin-like inhibitory substances from the species have been identified and their antimicrobial activities have been investigated for applications in food industries. [Cui et al. \(2020\)](#) reported that a bacteriocin

purified from *E. faecalis* strain CG-9 exhibited a wide range of antimicrobial activity against many food-borne spoilage and pathogenic bacteria (e.g., *E. coli*, *L. monocytogenes*, and *S. aureus*), and some yeast, and thus could have potential value for research and application value in the food industries. Additionally, the food-isolated *E. faecalis* OS13 was found to produce the enterocin OS13, which is a bacteriocin-like inhibitory substances comprising two peptides (enterocins OS13 α and OS13 β). This enterocin inhibited the growth of antibiotic-resistant nosocomial *E. faecalis* and *E. faecium* isolates; it was suggested to have potential for treating clinical infections with vancomycin-resistant and multidrug-resistant enterococci, and/or for preserving foods stored at low temperatures (El-Gendy et al., 2021).

Some strains belonging to the genus *Companilactobacillus* can produce bacteriocin (Menteş, Ercan, & Akçelik, 2007; P. Wang, Wu, et al., 2021), but no previous study reported a bacteriocin from *C. zhachilii*. Therefore, the bacteriocin and/or antimicrobial substances from the strains isolated in this study could be usefully developed for the food industries; in particular, *C. zhachilii* strain L113 may be considered a potential resource for the exploration and discovery of novel bacteriocin(s).

3.6. Metabolic analysis of LAB strains

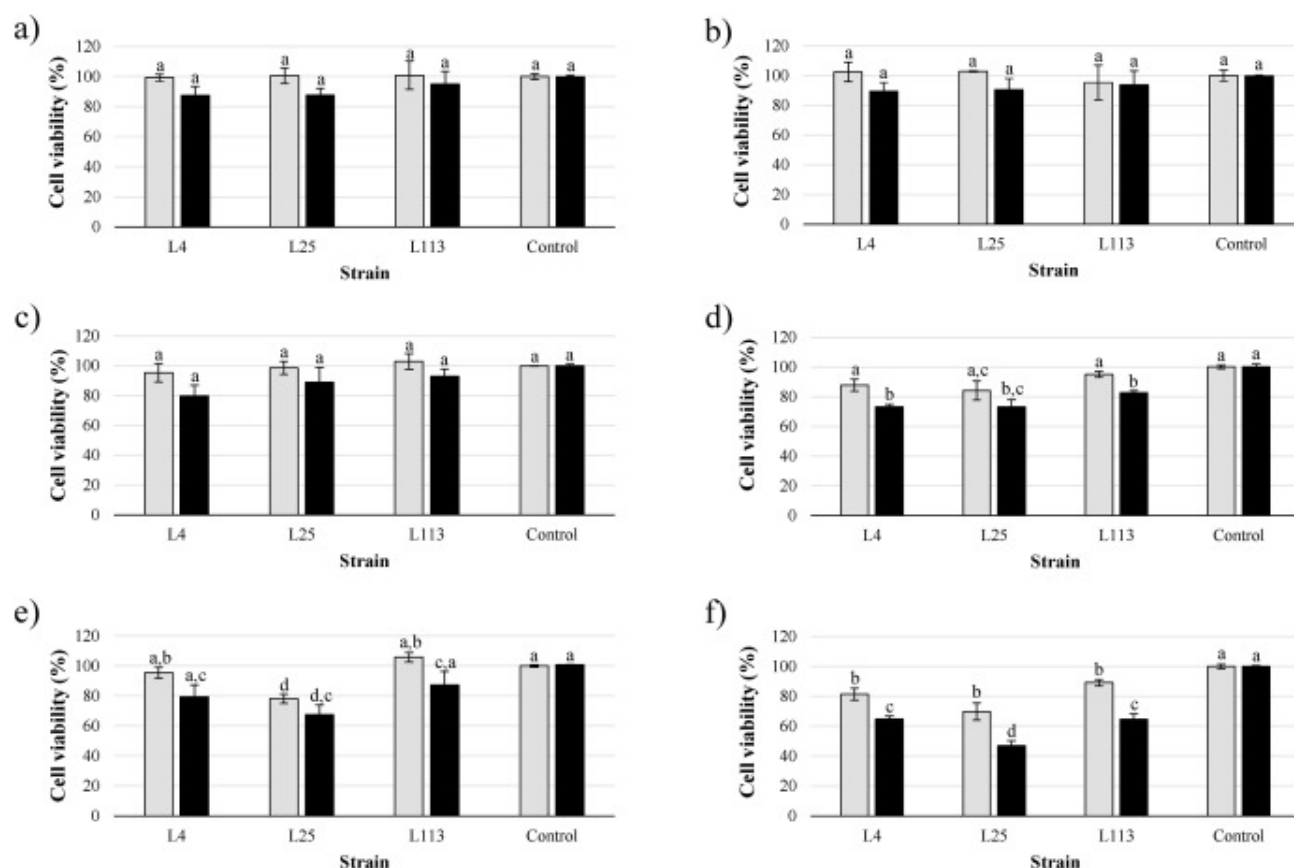
The metabolites produced by the isolated LAB strains were analyzed using GC-MS and major peaks were identified by comparison with the National Institute of Standards and Technology library. In the presence of purified LCMP in the media, trimethylsilyl derivatives of lactic acid and acetic acid were found in the culture media of all the LAB strains tested (Table 2 and Fig. S3). In addition, the culture media of *E. faecalis* strain L4 contained 2-hydroxyisocaproic acid, palmitic acid, and stearic acid, the culture media of *P. acidilactici* strain L25 contained benzoic acid, 3-phenyllactic acid, and palmitic acid, and the culture media of *C. zhachilii* strain L113 contained benzoic acid (Table 2). These results suggested that the acidity of the culture media of *E. faecalis* strain L4, *P. acidilactici* strain L25 and *C. zhachilii* strain L113 was due to the presence of metabolites, such as lactic acid, acetic acid, and other organic acids (Table 2).

Many bacteria including intestinal microflora have polysaccharide utilization loci (PULs), which are gene clusters encoding carbohydrate-active enzymes, glycan-binding proteins, transporters, sensors, and regulators required for binding, cleavage, and import of carbohydrates (Wardman, Bains, Rahfeld, & Withers, 2022). The presence of dietary polysaccharides that are considered prebiotics can affect the populations and diversity of intestinal microflora, including Bacteroidetes, Firmicutes, and Proteobacteria (Shen et al., 2023; Zhu et al., 2021). The polysaccharides can also affect their metabolic process for the conversion of polysaccharides into monosaccharides using the PULs, and the formation of various products in the gut, thereby enhancing the production of SCFAs, such as acetic acid, propionic acid, and butyric acid (Cheng, Hu, Geng, & Nie, 2022; Flint, Bayer, Rincon, Lamed, & White, 2008; Yang et al., 2021). During fermentation, bacteria consume the carbon source to produce pyruvate through glycolysis and then produce lactic acid with lactate dehydrogenase. However, during hetero-fermentation, LAB produce other organic acids, such as

acetic acid, formic acid, and ethanol through citric acid metabolism (Shimazu, Uehara, & Watanabe, 1985; Y. Wang, Wu, et al., 2021). In addition, some organic acids are produced by amino acid metabolism, for example, phenyllactic acid (PLA) is produced by phenylalanine catabolism, and 2-hydroxyisocaproic acid (HICA) is produced by leucine catabolism (Park et al., 2017; Prasuna, Mujahid, Sasikala, & Ramana, 2012; Sun et al., 2019). Although purified LCMP do not represent a preferred carbon source for LAB strain growth, the polysaccharides clearly induced increases in the amounts of organic acids and lactic acid in the media used to culture the LAB strains. These results suggested that purified LCMP stimulate the polysaccharide metabolism of bacterial cells to produce metabolites that can have direct and indirect effects on the gut environment, including the growth of other gut bacteria and host intestinal epithelial cells.

3.7. Inhibition of proliferation of HT-29 cells upon treatment with postbiotics derived from LAB strains

To investigate the effects that metabolites of isolated LAB strains may have on host intestinal cells, the viability of HT-29 colon cancer cells was determined after treatment with culture media from the LAB strains. The viabilities of HT-29 cells treated for 1–2 d with 5% LAB strain culture medium (Fig. 6a and c) and for 1 d with 15% LAB strain culture medium (Fig. 6b) were not significantly lower than the negative control. Compared to the negative control, there was also no significant change in viability when HT-29 cells were grown for 2 d with 15% culture media without purified LCMP from LAB strains (Fig. 6d, gray bars); however, viability of HT-29 cells was significantly decreased when treated for 2 d with 15% of LAB strain culture media with purified LCMP (Fig. 6d, black bars). Moreover, the viabilities of HT-29 cells treated for 3 d with 15% LAB strain culture media were significantly lower compared to the negative control, regardless of whether the LAB strains were grown with or without purified LCMP (Fig. 6f). Compared to the negative control, HT-29 cells treated with 15% culture media without purified LCMP from *E. faecalis* strain L4, *P. acidilactici* strain L25, and *C. zhachilii* strain L113 exhibited viabilities of 81.4%, 69.8%, and 89.1%, respectively, at 3 d (Fig. 6f, gray bars). When HT-29 cells were grown in LAB strain culture media with purified LCMP, the cell viabilities were only 64.6%, 46.8%, and 64.5%, respectively (Fig. 6f, black bars), which were approximately 0.79, 0.67 and 0.72 times their viabilities, respectively, when grown in culture media without purified LCMP. This result showed that the proliferation of HT-29 cells was inhibited by LAB strain culture media, and this effect was significantly stronger for culture media of LAB strains grown with purified LCMP compared to those grown without LCMP (Fig. 6f).



[Download: Download high-res image \(414KB\)](#)

[Download: Download full-size image](#)

Fig. 6. The effects of LAB strain culture media on the viability of HT-29 cells as measured by the MTS assay. LAB strains were cultured in the absence (gray bars) or presence (black bars) of purified LCMP, and the cell-free cultured media were applied to HT-29 cells at 5% or 15%. Cell viabilities were measured for HT-29 cells treated with 5% culture medium (a, c, and e) and 15% culture medium (b, d, and f) that have been obtained by the cultivation of each LAB strain for 1 (a and b), 2 (c and d), or 3 d (e and f). The control was fresh media that was never cultured with bacteria. Results are presented as averages (n = 3). Bars with the same letter did not differ significantly (p < 0.05).

For decades, it has been reported that metabolic secretions, exopolysaccharides, heat-killed cells, cell walls, and cytoplasmic extracts of lactobacilli have antitumor activities against HT-29 and Caco-2 epithelial colon carcinoma cell lines (Brady, Gallaher, & Busta, 2000; Li et al., 2022; Tuo et al., 2010; Wang et al., 2014; Yue et al., 2020). The anticancer effects have been multifaceted and involve multiple mechanisms. Lactobacilli compete with pathogenic bacteria for nutrients and synthesize substances, such as bacteriocins, lactic acid, and benzoic acid, that are antagonistic to pathogenic bacteria involved in the synthesis of tumor promoters and procarcinogens (Anantharaju et al., 2017; Choi, Patra, Kim, & Kang, 2018; Garbacz, 2022; McWilliam Leitch & Stewart, 2002; Sanni, Onilude, Ogunbanwo, & Smith, 1999). Also, SCFAs synthesized by lactobacilli reduce the intestinal pH and maintain intestinal homeostasis and normal physiological processes (Garbacz, 2022; van der Hee & Wells, 2021). Moreover, SCFAs are absorbed by various cells and tissues and influence their growth,

proliferation, and apoptosis (Hinnebusch, Meng, Wu, Archer, & Hodin, 2002; Tang, Chen, Jiang, & Nie, 2011). Thus, these reports imply that metabolites, such as lactic acid and benzoic acid, produced by *E. faecalis* strain L4, *P. acidilactici* strain L25, and *C. zhachilii* strain L113 can not only inhibit HT-29 proliferation but also aid in maintaining intestinal balance.

Previous studies also showed that palmitic acid and stearic acid have anticancer activities. Palmitic acid induced tumor cell apoptosis, interfered with the cancer cell cycle, and induced programmed cell autophagic death. Stearic acid induced morphological changes and apoptosis of breast cancer cells (Evans, Cowey, Siegal, & Hardy, 2009; Wang, Zhang, & Bao, 2023).

PLA, which is generally used as an antimicrobial compound, was shown to promote cell migration and invasion of cervical cancer cells, such as SiHa, HeLa, and C-33A, thus it may trigger cervical cancer cell motility (Li, Li, Sui, Wang, & Li, 2019). However, a recent study showed that PLA inhibited colon cancer caused by the adherence of *Fusobacterium nucleatum*, thus PLA has potential as a preventative and adjuvant treatment for colorectal cancer (Wu, Xu, Wang, Yang, & Xu, 2023).

The probiotic yeast *Saccharomyces boulardii* is actually a strain of *S. cerevisiae* and is known to produce many bioactive metabolites that have antioxidant, anticancer, and anti-inflammatory properties (Allahyari et al., 2020; Datta, Timson, & Annapure, 2017; Sougioultzis et al., 2006). An analysis of the metabolic differences between *S. boulardii* and *S. cerevisiae* showed that *S. boulardii* produced PLA and 2-hydroxyisocaproic acid, but *S. cerevisiae* did not (Fu et al., 2023). Based on the result, PLA and 2-hydroxyisocaproic acid were reported to have potential as anticancer agents (Pakbin et al., 2024).

As mentioned above, lactic acid, acetic acid, and metabolites, such as 2-hydroxyisocaproic acid, palmitic acid, stearic acid, benzoic acid, and/or 3-phenyllactic acid, were present in the culture media of *E. faecalis* strain L4, *P. acidilactici* strain L25, and *C. zhachilii* strain L113 grown in the presence of purified LCMP. Therefore, these previous studies support that inhibition of the proliferation of HT-29 cells by treatment with the cultured media is an effect of the metabolites in the culture media of the LAB strains. Additionally, further studies should examine to verify effective substances in the LAB strain culture media and find out the cell-death mechanism(s), including apoptosis and necrosis, that are triggered by treatment of the culture media.

4. Conclusions

To investigate the effect of purified LCMP on probiotic strains, LAB strains isolated from swine gut microflora were examined for aggregation, antibacterial activity, and metabolite production in growth medium supplemented with purified LCMP compared with medium without LCMP. In the presence of purified LCMP, *C. zhachilii* strain L113 showed auto- and co-aggregation with pathogenic bacteria, suggesting that it may survive by auto-aggregating and inhibiting the adhesion of pathogenic bacteria in the intestine. *P. acidilactici* strain L25 could survive at low pH and in the presence of bile salt, suggesting that it has a high potential to survive in the GIT of a host that eats

the strain as a food additive. Moreover, in the presence of purified LCMP, *P. acidilactici* strain L25 and *E. faecalis* strain L4 produced host-beneficial metabolites that could inhibit the proliferation of HT-29 cancer cells. Moreover, all of the strains were capable of inhibiting the growth of intestinal pathogenic bacteria. Taken together, these findings suggested that in the presence of purified LCMP, *E. faecalis* strain L4, *P. acidilactici* strain L25 and *C. zhachilii* strain L113, as well as other functionally similar probiotics to those, can maintain the intestinal microbial balance by inhibiting the growth of pathogens and produce bioactive metabolites that inhibit colon cancer cell proliferation. Therefore, purified LCMP is a suitable prebiotic that has positive effects on probiotic growth and metabolite production, and can be used as an effective synbiotic when used together with the isolated LAB strains.

Funding

This research was supported by the Korea Research Institute of Bioscience and Biotechnology (KRIBB) Research Initiative Program ([KGM5482423](#)).

CRedit authorship contribution statement

Ji Young Kang: Writing – review & editing, Writing – original draft, Visualization, Project administration, Formal analysis, Conceptualization. **Seonghun Kim:** Validation, Supervision, Methodology. **Jung-Mi Kim:** Writing – review & editing, Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

The following is the Supplementary data to this article.

 [Download: Download Acrobat PDF file \(563KB\)](#)

Multimedia component 1.

[Recommended articles](#)

Data availability

Data will be made available on request.

References

Allahyari et al., 2020 S. Allahyari, S.P. Dibazar, B. Pakbin, R. Mahmoudi, A. Farasat, A. Peymani, ..., N. Gheibi
Anticancer effect of probiotic *Saccharomyces boulardii* supernatant on human Caco-2 cells; an in vitro study

Carpathian Journal of Food Science and Technology, 12 (5) (2020), pp. 181-189,

[10.34302/crpjfst/2020.12.5.14](https://doi.org/10.34302/crpjfst/2020.12.5.14) ↗

[Google Scholar](#) ↗

Anantharaju et al., 2017 P.G. Anantharaju, B.D. Reddy, M.A. Padukudru, C.M. Kumari Chitturi, M.G. Vimalambike, S.V. Madhunapantula

Naturally occurring benzoic acid derivatives retard cancer cell growth by inhibiting histone deacetylases (HDAC)

Cancer Biology & Therapy, 18 (7) (2017), pp. 492-504, [10.1080/15384047.2017.1324374](https://doi.org/10.1080/15384047.2017.1324374) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Brady et al., 2000 L.J. Brady, D.D. Gallaher, F.F. Busta

The role of probiotic cultures in the prevention of colon cancer

The Journal of Nutrition, 130 (2) (2000), pp. 410S-414S, [10.1093/jn/130.2.410S](https://doi.org/10.1093/jn/130.2.410S) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

Busch and Stumm, 1968 P.L. Busch, W. Stumm

Chemical interactions in the aggregation of bacteria bioflocculation in waste treatment

Environmental Science and Technology, 2 (1) (1968), pp. 49-53, [10.1021/es60013a004](https://doi.org/10.1021/es60013a004) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Chen et al., 2006 N.-Y. Chen, T.-H. Hsu, F.-Y. Lin, H.-H. Lai, J.-Y. Wu

Effects on cytokine-stimulating activities of EPS from *Tremella mesenterica* with various carbon sources

Food Chemistry, 99 (1) (2006), pp. 92-97, [10.1016/j.foodchem.2005.07.023](https://doi.org/10.1016/j.foodchem.2005.07.023) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Chen et al., 2021 S. Chen, J. Wang, Q. Fang, N. Dong, Q. Fang, S.W. Cui, et al.

A polysaccharide from natural *Cordyceps sinensis* regulates the intestinal immunity and gut microbiota in mice with cyclophosphamide-induced intestinal injury

Food & Function, 12 (2021), pp. 6271-6282, [10.1039/D1FO00596K](https://doi.org/10.1039/D1FO00596K) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Cheng et al., 2022 J. Cheng, J. Hu, F. Geng, S. Nie

***Bacteroides* utilization for dietary polysaccharides and their beneficial effects on gut health**

Food Science and Human Wellness, 11 (5) (2022), pp. 1101-1110, [10.1016/j.fshw.2022.04.002](https://doi.org/10.1016/j.fshw.2022.04.002) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Chidambaram et al., 2022](#) S.B. Chidambaram, M.M. Essa, A. Rathipriya, M. Bishir, B. Ray, A.M. Mahalakshmi, ..., T.M. Monaghan

Gut dysbiosis, defective autophagy and altered immune responses in neurodegenerative diseases: Tales of a vicious cycle

Pharmacology & Therapeutics, 231 (2022), Article 107988, [10.1016/j.pharmthera.2021.107988](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Choi et al., 2018](#) A.-R. Choi, J.K. Patra, W.J. Kim, S.-S. Kang

Antagonistic activities and probiotic potential of lactic acid bacteria derived from a plant-based fermented food

Frontiers in Microbiology, 9 (2018), Article 365192, [10.3389/fmicb.2018.01963](#) ↗

[Google Scholar](#) ↗

[Choi and Woo, 2015](#) J.-M. Choi, G.-J. Woo

Transfer of tetracycline resistance genes with aggregation substance in food-borne *Enterococcus faecalis*

Current Microbiology, 70 (2015), pp. 476-484, [10.1007/s00284-014-0742-1](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Chu et al., 2023](#) J. Chu, S. Feng, C. Guo, B. Xue, K. He, L. Li

Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: A review

Biomedicine & Pharmacotherapy, 164 (2023), Article 114985, [10.1016/j.biopha.2023.114985](#) ↗

[Google Scholar](#) ↗

[Chung et al., 2016](#) W.S.F. Chung, A.W. Walker, P. Louis, J. Parkhill, J. Vermeiren, D. Bosscher, ..., H.J. Flint

Modulation of the human gut microbiota by dietary fibres occurs at the species level

BMC Biology, 14 (2016), pp. 1-13, [10.1186/s12915-015-0224-3](#) ↗

[Google Scholar](#) ↗

[Cronin et al., 2021](#) P. Cronin, S.A. Joyce, P.W. O'Toole, E.M. O'Connor

Dietary fibre modulates the gut microbiota

Nutrients, 13 (5) (2021), p. 1655, [10.3390/nu13051655](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Cui et al., 2020](#) G. Cui, C. Pan, P. Xu, Y. Li, L. Wang, B. Gong, ..., S. Huang

Purification and characterization of a novel bacteriocin produced by *Enterococcus faecalis* CG-9 from human saliva

Biotechnology & Biotechnological Equipment, 34 (1) (2020), pp. 1224-1233,

[10.1080/13102818.2020.1830714](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Datta et al., 2017](#) S. Datta, D.J. Timson, U.S. Annapure

Antioxidant properties and global metabolite screening of the probiotic yeast *Saccharomyces cerevisiae* var. *boulardii*

Journal of the Science of Food and Agriculture, 97 (9) (2017), pp. 3039-3049, [10.1002/jsfa.8147](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Debnath et al., 2021](#) N. Debnath, R. Kumar, A. Kumar, P.K. Mehta, A.K. Yadav

Gut-microbiota derived bioactive metabolites and their functions in host physiology

Biotechnology & Genetic Engineering Reviews, 37 (2) (2021), pp. 105-153,

[10.1080/02648725.2021.1989847](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[El-Gendy et al., 2021](#) A.O. El-Gendy, D.A. Brede, T.M. Essam, M.A. Amin, S.H. Ahmed, H. Holo, ..., Y.I. Shamikh

Purification and characterization of bacteriocins-like inhibitory substances from food isolated *Enterococcus faecalis* OS13 with activity against nosocomial enterococci

Scientific Reports, 11 (1) (2021), p. 3795, [10.1038/s41598-021-83357-z](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Evans et al., 2009](#) L.M. Evans, S.L. Cowey, G.P. Siegal, R.W. Hardy

Stearate preferentially induces apoptosis in human breast cancer cells

Nutrition and Cancer, 61 (5) (2009), pp. 746-753, [10.1080/01635580902825597](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Flint et al., 2008](#) H.J. Flint, E.A. Bayer, M.T. Rincon, R. Lamed, B.A. White

Polysaccharide utilization by gut bacteria: Potential for new insights from genomic analysis

Nature Reviews Microbiology, 6 (2) (2008), pp. 121-131, [10.1038/nrmicro1817](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Françoise, 2010](#) L. Françoise

Occurrence and role of lactic acid bacteria in seafood products

Food Microbiology, 27 (6) (2010), pp. 698-709, [10.1016/j.fm.2010.05.016](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Fu et al., 2023](#) J. Fu, J. Liu, X. Wen, G. Zhang, J. Cai, Z. Qiao, ..., L. Li

Unique probiotic properties and bioactive metabolites of *Saccharomyces boulardii*

Probiotics and Antimicrobial Proteins, 15 (4) (2023), pp. 967-982, [10.1007/s12602-022-09953-1](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Fugaban et al., 2022](#) J.I.I. Fugaban, J.E. Vazquez Bucheli, Y.J. Park, D.H. Suh, E.S. Jung, B.D. G.d.M. Franco, ...,

S.D. Todorov

Antimicrobial properties of *Pediococcus acidilactici* and *Pediococcus pentosaceus* isolated from silage

Journal of Applied Microbiology, 132 (1) (2022), pp. 311-330, [10.1111/jam.15205](https://doi.org/10.1111/jam.15205) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Garbacz, 2022](#) K. Garbacz

Anticancer activity of lactic acid bacteria

Seminars in Cancer Biology, 86 (3) (2022), pp. 356-366, [10.1016/j.semcancer.2021.12.013](https://doi.org/10.1016/j.semcancer.2021.12.013) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Goh et al., 2022](#) K.S. Goh, Z.J. Ng, M. Halim, S.N. Oslan, S.N.H. Oslan, J.S. Tan

A comprehensive review on the anticancer potential of bacteriocin: Preclinical and clinical studies

International Journal of Peptide Research and Therapeutics, 28 (2) (2022), p. 75, [10.1007/s10989-022-10386-1](https://doi.org/10.1007/s10989-022-10386-1) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Gold and Zhu, 2022](#) A. Gold, J. Zhu

Not just a gut feeling: A deep exploration of functional bacterial metabolites that can modulate host health

Gut Microbes, 14 (1) (2022), Article 2125734, [10.1080/19490976.2022.2125734](https://doi.org/10.1080/19490976.2022.2125734) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Gross et al., 2008](#) G. Gross, J. van Der Meulen, J. Snel, R. van Der Meer, M. Kleerebezem, T.A. Niewold, ..., M.A. Smits

Mannose-specific interaction of *Lactobacillus plantarum* with porcine jejunal epithelium

FEMS Immunology and Medical Microbiology, 54 (2) (2008), pp. 215-223, [10.1111/j.1574-695X.2008.00466.x](https://doi.org/10.1111/j.1574-695X.2008.00466.x) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Guo et al., 2017](#) M.Q. Guo, X. Hu, C. Wang, L. Ai

Polysaccharides: Structure and solubility

Solubility of polysaccharides, 2 (2017), pp. 8-21, [10.5772/intechopen.71570](https://doi.org/10.5772/intechopen.71570) ↗

[Google Scholar](#) ↗

[Hinnebusch et al., 2002](#) B.F. Hinnebusch, S. Meng, J.T. Wu, S.Y. Archer, R.A. Hodin

The effects of short-chain fatty acids on human colon cancer cell phenotype are associated with histone hyperacetylation

The Journal of Nutrition, 132 (5) (2002), pp. 1012-1017, [10.1093/jn/132.5.1012](https://doi.org/10.1093/jn/132.5.1012) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Hojjati et al., 2020 M. Hojjati, B.A. Behabehani, F. Falah

Aggregation, adherence, anti-adhesion and antagonistic activity properties relating to surface charge of probiotic *Lactobacillus brevis* gp104 against *Staphylococcus aureus*

Microbial Pathogenesis, 147 (2020), Article 104420, [10.1016/j.micpath.2020.104420](https://doi.org/10.1016/j.micpath.2020.104420) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Hytönen et al., 2003 J. Hytönen, S. Haataja, J. Finne

Streptococcus pyogenes glycoprotein-binding strepadhesin activity is mediated by a surface-associated carbohydrate-degrading enzyme, pullulanase

Infection and Immunity, 71 (2) (2003), pp. 784-793, [10.1128/iai.71.2.784-793.2003](https://doi.org/10.1128/iai.71.2.784-793.2003) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Isenring et al., 2021 J. Isenring, A. Geirnaert, C. Lacroix, M.J. Stevens

Bistable auto-aggregation phenotype in *Lactiplantibacillus plantarum* emerges after cultivation in in vitro colonic microbiota

BMC Microbiology, 21 (2021), pp. 1-13, [10.1186/s12866-021-02331-x](https://doi.org/10.1186/s12866-021-02331-x) ↗

[Google Scholar](#) ↗

Johannsen and Prescott, 2009 H. Johannsen, S. Prescott

Practical prebiotics, probiotics and synbiotics for allergists: How useful are they?

Clinical and Experimental Allergy, 39 (12) (2009), pp. 1801-1814, [10.1111/j.1365-2222.2009.03368.x](https://doi.org/10.1111/j.1365-2222.2009.03368.x) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Juntarachot et al., 2023 N. Juntarachot, S. Sunpaweravong, A. Kaewdech, M. Wongsuwanlert, P. Ruangsri, N. Pahumunto, et al.

Characterization of adhesion, anti-adhesion, co-aggregation, and hydrophobicity of *Helicobacter pylori* and probiotic strains

Journal of Taibah University Medical Sciences, 18 (5) (2023), p. 1048, [10.1016/j.jtumed.2023.02.017](https://doi.org/10.1016/j.jtumed.2023.02.017) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Kamada et al., 2013 N. Kamada, G.Y. Chen, N. Inohara, G. Núñez

Control of pathogens and pathobionts by the gut microbiota

Nature Immunology, 14 (7) (2013), pp. 685-690, [10.1038/ni.2608](https://doi.org/10.1038/ni.2608) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Kang et al., 2022 J.Y. Kang, B. Lee, C.H. Kim, J.H. Choi, M.-S. Kim

Enhancing the prebiotic and antioxidant effects of exopolysaccharides derived from *Cordyceps militaris* by enzyme-digestion

Lebensmittel-Wissenschaft und -Technologie, 167 (2022), Article 113830, [10.1016/j.lwt.2022.113830](https://doi.org/10.1016/j.lwt.2022.113830) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Kang et al., 2023](#) J.Y. Kang, H.-Y. Song, J.-M. Kim

Agarolytic pathway in the newly isolated *Aquimarina* sp. bacterial strain ERC-38 and characterization of a putative β -agarase

Marine Biotechnology, 25 (2) (2023), pp. 314-327, [10.1007/s10126-023-10206-7](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Kasimin et al., 2022](#) M.E. Kasimin, S. Shamsuddin, A.M. Molujin, M.K. Sabullah, J.A. Gansau, R. Jawan

Enterocin: Promising biopreservative produced by *Enterococcus* sp

Microorganisms, 10 (4) (2022), p. 684, [10.3390/microorganisms10040684](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Kim et al., 2006](#) Y.O. Kim, H.W. Park, J.H. Kim, J.Y. Lee, S.H. Moon, C.S. Shin

Anti-cancer effect and structural characterization of endo-polysaccharide from cultivated mycelia of *Inonotus obliquus*

Life Sciences, 79 (1) (2006), pp. 72-80, [10.1016/j.lfs.2005.12.047](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Kumar and Sharma, 2022](#) K. Kumar, R. Sharma

Role of probiotics in treatment of gut-related diseases

Prebiotics and Probiotics in Disease Regulation and Management (2022), pp. 1-26,

[10.1002/9781394167227.ch1](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Li et al., 2022](#) F. Li, X. Jiao, J. Zhao, X. Liao, Y. Wei, Q. Li

Antitumor mechanisms of an exopolysaccharide from *Lactobacillus fermentum* on HT-29 cells and HT-29 tumor-bearing mice

International Journal of Biological Macromolecules, 209 (2022), pp. 552-562,

[10.1016/j.ijbiomac.2022.04.023](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Li et al., 2019](#) C. Li, Y. Li, L. Sui, J. Wang, F. Li

Phenyllactic acid promotes cell migration and invasion in cervical cancer via IKK/NF- κ B-mediated MMP-9 activation

Cancer Cell International, 19 (2019), pp. 1-12, [10.1186/s12935-019-0965-0](#) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

[Liu et al., 2014](#) W. Liu, H. Pang, H. Zhang, Y. Cai

Biodiversity of lactic acid bacteria

Lactic acid bacteria: Fundamentals and practice (2014), pp. 103-203, [10.1007/978-94-017-8841-0_2](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Liu et al., 2024](#) W.-X. Liu, J.-J. Wang, X.-K. Xiao, C.-R. Chen, X. Lu, X.-Y. Zhang, ..., F. Wang

Antimicrobial effects and metabolomics analysis of cell-free supernatant produced by *Pediococcus acidilactici* LWX 401 isolated from Yunnan traditional pickles

Lwt, 191 (2024), Article 115626, [10.1016/j.lwt.2023.115626](https://doi.org/10.1016/j.lwt.2023.115626) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Malik et al., 2013 S. Malik, M.I. Petrova, I.J. Claes, T.L. Verhoeven, P. Busschaert, M. Vaneechoutte, ..., S. Lebeer

The highly autoaggregative and adhesive phenotype of the vaginal *Lactobacillus plantarum* strain CMPG5300 is sortase dependent

Applied and Environmental Microbiology, 79 (15) (2013), pp. 4576-4585, [10.1128/AEM.00926-13](https://doi.org/10.1128/AEM.00926-13) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Mao et al., 2020 Y.-H. Mao, A.-X. Song, L.-Q. Li, K.-C. Siu, Z.-P. Yao, J.-Y. Wu

Effects of exopolysaccharide fractions with different molecular weights and compositions on fecal microflora during in vitro fermentation

International Journal of Biological Macromolecules, 144 (2020), pp. 76-84,

[10.1016/j.ijbiomac.2019.12.072](https://doi.org/10.1016/j.ijbiomac.2019.12.072) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Maske et al., 2021 B.L. Maske, G.V. de Melo Pereira, A.d.S. Vale, D.P. de Carvalho Neto, S.G. Karp, J.A. Viesser, ..., C.R. Soccol

A review on enzyme-producing lactobacilli associated with the human digestive process: From metabolism to application

Enzyme and Microbial Technology, 149 (2021), Article 109836, [10.1016/j.enzmictec.2021.109836](https://doi.org/10.1016/j.enzmictec.2021.109836) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Masood et al., 2011 M.I. Masood, M.I. Qadir, J.H. Shirazi, I.U. Khan

Beneficial effects of lactic acid bacteria on human beings

Critical Reviews in Microbiology, 37 (1) (2011), pp. 91-98, [10.3109/1040841X.2010.536522](https://doi.org/10.3109/1040841X.2010.536522) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Masuko et al., 2005 T. Masuko, A. Minami, N. Iwasaki, T. Majima, S.-I. Nishimura, Y.C. Lee

Carbohydrate analysis by a phenol–sulfuric acid method in microplate format

Analytical Biochemistry, 339 (1) (2005), pp. 69-72, [10.1016/j.ab.2004.12.001](https://doi.org/10.1016/j.ab.2004.12.001) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

McWilliam Leitch and Stewart, 2002 E. McWilliam Leitch, C. Stewart

Escherichia coli O157 and non-O157 isolates are more susceptible to L-lactate than to D-lactate

Applied and Environmental Microbiology, 68 (9) (2002), pp. 4676-4678, [10.1128/AEM.68.9.4676-4678.2002](https://doi.org/10.1128/AEM.68.9.4676-4678.2002) ↗

[Google Scholar](#) ↗

[Menteş et al., 2007](#) Ö. Menteş, R. Ercan, M. Akçelik

Inhibitor activities of two *Lactobacillus* strains, isolated from sourdough, against rope-forming *Bacillus* strains

Food Control, 18 (4) (2007), pp. 359-363, [10.1016/j.foodcont.2005.10.020](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Møller et al., 2017](#) M.S. Møller, Y.J. Goh, K.B. Rasmussen, W. Cypryk, H.U. Celebioglu, T.R. Klaenhammer, ..., M. Abou Hachem

An extracellular cell-attached pullulanase confers branched α -glucan utilization in human gut *Lactobacillus acidophilus*

Applied and Environmental Microbiology, 83 (12) (2017), Article e00402, [10.1128/AEM.00402-17](#) ↗
00417

[Google Scholar](#) ↗

[Mwangi et al., 2022](#) R.W. Mwangi, J.M. Macharia, I.N. Wagara, R.L. Bence

The antioxidant potential of different edible and medicinal mushrooms

Biomedicine & Pharmacotherapy, 147 (2022), Article 112621, [10.1016/j.biopha.2022.112621](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Nisha and Satyanarayana, 2013](#) M. Nisha, T. Satyanarayana

Recombinant bacterial amylopullulanases: Developments and perspectives

Bioengineered, 4 (6) (2013), pp. 388-400, [10.4161/bioe.24629](#) ↗

[Google Scholar](#) ↗

[Nwoko and Okeke, 2021](#) E.-s. Q. Nwoko, I.N. Okeke

Bacteria autoaggregation: How and why bacteria stick together

Biochemical Society Transactions, 49 (3) (2021), pp. 1147-1157, [10.1042/BST20200718](#) ↗

[View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Pakbin et al., 2024](#) B. Pakbin, S. Allahyari, S.P. Dibazar, A. Peymani, M.K. Haghverdi, K. Taherkhani, ..., R. Mahmoudi

Anticancer properties of *Saccharomyces boulardii* metabolite against colon cancer cells

Probiotics and Antimicrobial Proteins, 16 (1) (2024), pp. 224-232, [10.1007/s12602-022-10030-w](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Papagianni and Anastasiadou, 2009](#) M. Papagianni, S. Anastasiadou

Pediocins: The bacteriocins of Pediococci. Sources, production, properties and applications

Microbial Cell Factories, 8 (2009), pp. 1-16, [10.1186/1475-2859-8-3](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Park et al., 2017](#) B. Park, H. Hwang, J.Y. Chang, S.W. Hong, S.H. Lee, M.Y. Jung, ..., J.-H. Lee

Identification of 2-hydroxyisocaproic acid production in lactic acid bacteria and evaluation of microbial dynamics during kimchi ripening

Scientific Reports, 7 (1) (2017), Article 10904, [10.1038/s41598-017-10948-0](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Peluzio et al., 2021](#) M.d.C.G. Peluzio, J.A. Martinez, F.I. Milagro

Postbiotics: Metabolites and mechanisms involved in microbiota-host interactions

Trends in Food Science & Technology, 108 (2021), pp. 11-26, [10.1016/j.tifs.2020.12.004](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Prasuna et al., 2012](#) M.L. Prasuna, M. Mujahid, C. Sasikala, C.V. Ramana

L-Phenylalanine catabolism and L-phenyllactic acid production by a phototrophic bacterium, *Rubrivivax benzoatilyticus* JA2

Microbiological Research, 167 (9) (2012), pp. 526-531, [10.1016/j.micres.2012.03.001](#) ↗

[Google Scholar](#) ↗

[Rahmatabadi et al., 2017](#) S.S. Rahmatabadi, I. Sadeghian, N. Nezafat, M. Negahdaripour, N.

Hajighahramani, S. Hemmati, et al.

In silico investigation of pullulanase enzymes from various *Bacillus* species

Current Proteomics, 14 (3) (2017), pp. 175-185, [10.2174/1570164614666170306164830](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Ramos et al., 2020](#) S. Ramos, V. Silva, M.d.L.E. Dapkevicius, G. Igrejas, P. Poeta

Enterococci, from harmless bacteria to a pathogen

Microorganisms, 8 (8) (2020), p. 1118, [10.3390/microorganisms8081118](#) ↗

[Google Scholar](#) ↗

[Rathore et al., 2017](#) H. Rathore, S. Prasad, S. Sharma

Mushroom nutraceuticals for improved nutrition and better human health: A review

PharmaNutrition, 5 (2) (2017), pp. 35-46, [10.1016/j.phanu.2017.02.001](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Sanni et al., 1999](#) A. Sanni, A. Onilude, S. Ogunbanwo, S. Smith

Antagonistic activity of bacteriocin produced by *Lactobacillus* species from ogi, an indigenous fermented food

Journal of Basic Microbiology: An International Journal on Biochemistry, Physiology, Genetics, Morphology, and Ecology of Microorganisms, 39 (3) (1999), pp. 189-195, [10.1002/\(SICI\)1521-4028\(199906\)39:3<189::AID-JOBM189>3.0.CO;2-R](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Shen et al., 2023](#) L. Shen, Y. Shen, L. You, Y. Zhang, Z. Su, G. Peng, ..., S. Cao

Pueraria lobata polysaccharides alleviate neonatal calf diarrhea by modulating gut microbiota and metabolites

Frontiers in Veterinary Science, 9 (2023), Article 1024392, [10.3389/fvets.2022.1024392](https://doi.org/10.3389/fvets.2022.1024392) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Shimazu et al., 1985](#) Y. Shimazu, M. Uehara, M. Watanabe

Transformation of citric acid to acetic acid, acetoin and diacetyl by wine making lactic acid bacteria

Agricultural and Biological Chemistry, 49 (7) (1985), pp. 2147-2157, [10.1080/00021369.1985.10867041](https://doi.org/10.1080/00021369.1985.10867041) ↗

[Google Scholar ↗](#)

[Singh et al., 2018](#) K.S. Singh, S. Kumar, A.K. Mohanty, S. Grover, J.K. Kaushik

Mechanistic insights into the host-microbe interaction and pathogen exclusion mediated by the Mucus-binding protein of *Lactobacillus plantarum*

Scientific Reports, 8 (1) (2018), Article 14198, [10.1038/s41598-018-32417-y](https://doi.org/10.1038/s41598-018-32417-y) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Sougioultzis et al., 2006](#) S. Sougioultzis, S. Simeonidis, K.R. Bhaskar, X. Chen, P.M. Anton, S. Keates, ..., C.P. Kelly

Saccharomyces boulardii produces a soluble anti-inflammatory factor that inhibits NF- κ B-mediated IL-8 gene expression

Biochemical and Biophysical Research Communications, 343 (1) (2006), pp. 69-76,

[10.1016/j.bbrc.2006.02.080](https://doi.org/10.1016/j.bbrc.2006.02.080) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Stewart, 2022](#) G.C. Stewart

Streptococcus and *Enterococcus*

Veterinary Microbiology (2022), pp. 240-251, [10.1002/9781119650836.ch26](https://doi.org/10.1002/9781119650836.ch26) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Sugrue et al., 2024](#) I. Sugrue, R.P. Ross, C. Hill

Bacteriocin diversity, function, discovery and application as antimicrobials

Nature Reviews Microbiology (2024), pp. 1-16, [10.1038/s41579-024-01045-x](https://doi.org/10.1038/s41579-024-01045-x) ↗

[Google Scholar ↗](#)

[Sun et al., 2019](#) D. Sun, H. Li, D. Song, L. Zhang, X. Zhao, X. Xu

Genome, transcriptome and fermentation analyses of *Lactobacillus plantarum* LY-78 provide new insights into the mechanism of phenyllactate biosynthesis in lactic acid bacteria

Biochemical and Biophysical Research Communications, 519 (2) (2019), pp. 351-357,

[10.1016/j.bbrc.2019.09.011](https://doi.org/10.1016/j.bbrc.2019.09.011) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Tang et al., 2011](#) Y. Tang, Y. Chen, H. Jiang, D. Nie

The role of short-chain fatty acids in orchestrating two types of programmed cell death in colon cancer

Autophagy, 7 (2) (2011), pp. 235-237, [10.4161/auto.7.2.14277](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Trunk et al., 2018](#) T. Trunk, H.S. Khalil, J.C. Leo

Bacterial autoaggregation

AIMS microbiology, 4 (1) (2018), pp. 140-164, [10.3934/microbiol.2018.1.140](#) ↗

[Google Scholar](#) ↗

[Tuo et al., 2010](#) Y. Tuo, L. Zhang, H. Yi, Y. Zhang, W. Zhang, X. Han, ..., S. Wang

Antiproliferative effect of wild *Lactobacillus* strains isolated from fermented foods on HT-29 cells

Journal of Dairy Science, 93 (6) (2010), pp. 2362-2366, [10.3168/jds.2010-3069](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Uccello et al., 2012](#) M. Uccello, G. Malaguarnera, F. Basile, V. D'agata, M. Malaguarnera, G. Bertino, ..., A. Biondi

Potential role of probiotics on colorectal cancer prevention

BMC Surgery, 12 (2012), pp. 1-8, [10.1186/1471-2482-12-S1-S35](#) ↗

[Google Scholar](#) ↗

[van der Hee and Wells, 2021](#) B. van der Hee, J.M. Wells

Microbial regulation of host physiology by short-chain fatty acids

Trends in Microbiology, 29 (8) (2021), pp. 700-712, [10.1016/j.tim.2021.02.001](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Wang et al., 2021](#) P. Wang, T. Wang, M. Ismael, X. Wang, Y. Yi, X. Lü

Development of an electroporation method and expression patterns of bacteriocin-encoding genes in *Companilactobacillus crustorum* MN047

Food Bioscience, 44 (2021), Article 101420, [10.1016/j.fbio.2021.101420](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Wang et al., 2017](#) Q. Wang, F. Wang, Z. Xu, Z. Ding

Bioactive mushroom polysaccharides: A review on monosaccharide composition, biosynthesis and regulation

Molecules, 22 (6) (2017), p. 955, [10.3390/molecules22060955](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Wang et al., 2021](#) Y. Wang, J. Wu, M. Lv, Z. Shao, M. Hungwe, J. Wang, ..., W. Geng

Metabolism characteristics of lactic acid bacteria and the expanding applications in

food industry

Frontiers in Bioengineering and Biotechnology, 9 (2021), Article 612285, [10.3389/fbioe.2021.612285](https://doi.org/10.3389/fbioe.2021.612285) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Wang et al., 2023](#) X. Wang, C. Zhang, N. Bao

Molecular mechanism of palmitic acid and its derivatives in tumor progression

Frontiers in Oncology, 13 (2023), Article 1224125, [10.3389/fonc.2023.1224125](https://doi.org/10.3389/fonc.2023.1224125) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Wang et al., 2014](#) S.-M. Wang, L.-W. Zhang, R.-B. Fan, X. Han, H.-X. Yi, L.-L. Zhang, ..., N. Shigwedha

Induction of HT-29 cells apoptosis by lactobacilli isolated from fermented products

Research in Microbiology, 165 (3) (2014), pp. 202-214, [10.1016/j.resmic.2014.02.004](https://doi.org/10.1016/j.resmic.2014.02.004) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Wardman et al., 2022](#) J.F. Wardman, R.K. Bains, P. Rahfeld, S.G. Withers

Carbohydrate-active enzymes (CAZymes) in the gut microbiome

Nature Reviews Microbiology, 20 (9) (2022), pp. 542-556, [10.1038/s41579-022-00712-1](https://doi.org/10.1038/s41579-022-00712-1) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Wu et al., 2023](#) X. Wu, J. Xu, D. Wang, X. Yang, X. Xu

A new anti-colon cancer tumor pathway of Phenyllactic acid by reducing adhesion of *Fusobacterium nucleatum*

Food Science and Technology (2023), [10.1590/fst.123222](https://doi.org/10.1590/fst.123222) ↗

[Google Scholar ↗](#)

[Xiang et al., 2012](#) Y. Xiang, X. Xu, J. Li

Chemical properties and antioxidant activity of exopolysaccharides fractions from mycelial culture of *Inonotus obliquus* in a ground corn stover medium

Food Chemistry, 134 (4) (2012), pp. 1899-1905, [10.1016/j.foodchem.2012.03.121](https://doi.org/10.1016/j.foodchem.2012.03.121) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Yadav et al., 2013](#) A.K. Yadav, A. Tyagi, J.K. Kaushik, A.C. Saklani, S. Grover, V.K. Batish

Role of surface layer collagen binding protein from indigenous *Lactobacillus plantarum* 91 in adhesion and its anti-adhesion potential against gut pathogen

Microbiological Research, 168 (10) (2013), pp. 639-645, [10.1016/j.micres.2013.05.003](https://doi.org/10.1016/j.micres.2013.05.003) ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Yang et al., 2021](#) M. Yang, F. Meng, W. Gu, L. Fu, F. Zhang, F. Li, ..., J. Yu

Influence of polysaccharides from *Polygonatum kingianum* on short-chain fatty acid production and quorum sensing in *Lactobacillus faecis*

Frontiers in Microbiology, 12 (2021), Article 758870, [10.3389/fmicb.2021.758870](https://doi.org/10.3389/fmicb.2021.758870) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Yoon et al., 2017](#) S.-H. Yoon, S.-M. Ha, S. Kwon, J. Lim, Y. Kim, H. Seo, *et al.*

Introducing EzBioCloud: A taxonomically united database of 16S rRNA gene sequences and whole-genome assemblies

International Journal of Systematic and Evolutionary Microbiology, 67 (5) (2017), p. 1613,

[10.1099/ijsem.0.001755](https://doi.org/10.1099/ijsem.0.001755) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Yue et al., 2020](#) Y.-C. Yue, B.-Y. Yang, J. Lu, S.-W. Zhang, L. Liu, K. Nassar, ..., J.-P. Lv

Metabolite secretions of *Lactobacillus plantarum* YYC-3 may inhibit colon cancer cell metastasis by suppressing the VEGF-MMP2/9 signaling pathway

Microbial Cell Factories, 19 (2020), pp. 1-12, [10.1186/s12934-020-01466-2](https://doi.org/10.1186/s12934-020-01466-2) ↗

[Google Scholar](#) ↗

[Zawistowska-Rojek et al., 2022](#) A. Zawistowska-Rojek, A. Kośmider, K. Stępień, S. Tyski

Adhesion and aggregation properties of *Lactobacillaceae* strains as protection ways against enteropathogenic bacteria

Archives of Microbiology, 204 (5) (2022), p. 285, [10.1007/s00203-022-02889-8](https://doi.org/10.1007/s00203-022-02889-8) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Zhu et al., 2021](#) K. Zhu, H. Fan, S. Zeng, S. Nie, Y. Zhang, L. Tan, ..., G. Wu

Polysaccharide from *Artocarpus heterophyllus* Lam.(jackfruit) pulp modulates gut microbiota composition and improves short-chain fatty acids production

Food Chemistry, 364 (2021), Article 130434, [10.1016/j.foodchem.2021.130434](https://doi.org/10.1016/j.foodchem.2021.130434) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Cited by (1)

[Emerging Nonthermal Technologies for the Production of Postbiotics](#) ↗

2025, Comprehensive Reviews in Food Science and Food Safety

© 2024 The Authors. Published by Elsevier Ltd.

similar technologies. For all open access content, the relevant licensing terms apply.